

A Verification Note on an Exponential Linear Program for Expected Average Nash Welfare under Interim Envy-Freeness

Verification note

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Abstract

This note verifies a proposed linear-programming formulation for maximizing expected average Nash welfare among interim envy-free lotteries in matching allocation instances. The mathematical core of the formulation is correct: after multiplying by the probability of the conditioning event, the interim envy-freeness constraints become linear inequalities in the probabilities assigned to perfect matchings. Consequently, if an interim envy-free lottery exists, an optimum for expected average Nash welfare is obtained by a finite linear program with one variable for each perfect matching. This gives an exact exponential-time algorithm and a polynomial upper bound on the support size of some optimal lottery. However, this is not a solution to the computational open problem of Caragiannis, Kanellopoulos, and Kyropoulou. Their framework already contains the exponential matching LP; the unresolved issue is whether the expected average Nash objective admits a polynomial-time algorithm, a suitable separation oracle, or a complexity classification. Thus the status of the result proved here is: correct but only modest partial progress/restatement, not a resolution of the authors' problem.

1 Introduction and status of the result

Caragiannis, Kanellopoulos, and Kyropoulou introduced and studied interim envy-free (iEF) lotteries over allocations, with particular attention to matching allocation instances [1]. In a matching instance, there are equally many agents and items, and each deterministic allocation is a perfect matching. Their paper gives polynomial-time algorithms for several welfare objectives under iEF constraints, including expected utilitarian welfare, expected egalitarian welfare, and expected log-Nash welfare. They explicitly distinguish expected log-Nash welfare from expected average Nash welfare for lotteries and leave the latter as elusive. The corresponding open-problem formulation asks for an algorithm, or a computational-complexity classification, for computing an iEF lottery maximizing

$$\mathbb{E}_{\mu \sim Q} [\text{avN}(\mu)], \quad \text{avN}(\mu) = \left(\prod_{i \in N} v_i(\mu(i)) \right)^{1/n}.$$

See also the open-problem page [2].

The purpose of this note is deliberately narrower than that open problem. We prove that the natural exponential-size primal LP over matchings is exactly correct for expected average Nash welfare. This verifies the main mathematical observation in the proposed solution attempt, while also clarifying its limitation.

Status. The result below is not a polynomial-time algorithm, not a hardness proof, and not a complexity classification. It is an exact exponential-time formulation. Therefore

it should be regarded as a correct finite-dimensional restatement, or at most a modest partial result, rather than a solution of the authors' open problem.

The main technical point is simple but important: the iEF condition is conditional, but it is conditional on an event whose probability is itself linear in the lottery. Multiplying through by that probability removes the conditional expectation without changing the constraint, because the constraint is imposed only when the conditioning event has positive probability and is vacuous otherwise.

2 Model

Let N be a set of n agents and let I be a set of n indivisible items. Agent $i \in N$ has a nonnegative additive valuation v_i , so the value of a single item $g \in I$ is denoted $v_i(g) \geq 0$. Since each agent receives exactly one item in a matching instance, only these singleton values are needed.

A deterministic matching is a bijection

$$\mu : N \rightarrow I.$$

Let $\mathcal{M}$ be the set of all deterministic matchings. Thus $|\mathcal{M}| = n!$. A lottery over matchings is a probability vector

$$q = (q_\mu)_{\mu \in \mathcal{M}}, \quad q_\mu \geq 0, \quad \sum_{\mu \in \mathcal{M}} q_\mu = 1.$$

Equivalently, a random matching μ is drawn according to q .

Definition 1 (Interim envy-freeness). *A lottery q is interim envy-free if, for every ordered pair of agents $i, j \in N$ and every item $g \in I$ satisfying*

$$\mathbb{P}_{\mu \sim q}[\mu(i) = g] > 0,$$

one has

$$v_i(g) \geq \mathbb{E}_{\mu \sim q} [v_i(\mu(j)) \mid \mu(i) = g]. \quad (1)$$

For a deterministic matching μ , define its average Nash welfare by

$$\text{avN}(\mu) = \left(\prod_{i \in N} v_i(\mu(i)) \right)^{1/n}. \quad (2)$$

If one of the selected singleton values is zero, then $\text{avN}(\mu) = 0$. For a lottery q , the objective considered here is the expected average Nash welfare

$$\mathbb{E}_{\mu \sim q}[\text{avN}(\mu)] = \sum_{\mu \in \mathcal{M}} q_\mu \text{avN}(\mu). \quad (3)$$

Although $\text{avN}(\mu)$ is nonlinear in the utilities of a deterministic matching, the expectation in (3) is linear in the lottery probabilities q_μ .

We shall also use the following caveat, which is sometimes easy to overlook.

Proposition 2 (iEF feasibility is not automatic). *There are matching instances for which no iEF lottery exists.*

Proof. Take two agents 1, 2 and two items a, b . Let both agents have values

$$v_1(a) = v_2(a) = 1, \quad v_1(b) = v_2(b) = 0.$$

There are two perfect matchings. Let α be the probability of the matching assigning a to agent 1 and b to agent 2. Then $1 - \alpha$ is the probability of the matching assigning b to agent 1 and a to agent 2.

If $\alpha > 0$, then with positive probability agent 2 receives b . Conditional on this event, agent 1 receives a , so iEF for agent 2 against agent 1 would require

$$v_2(b) \geq v_2(a),$$

namely $0 \geq 1$, impossible. Therefore iEF requires $\alpha = 0$.

If $1 - \alpha > 0$, then with positive probability agent 1 receives b . Conditional on this event, agent 2 receives a , so iEF for agent 1 against agent 2 would require $0 \geq 1$, again impossible. Therefore iEF requires $\alpha = 1$.

The two requirements $\alpha = 0$ and $\alpha = 1$ are incompatible. Hence no iEF lottery exists in this instance. $\square$

Thus an algorithm must either return an optimal iEF lottery or correctly report infeasibility.

3 Linearizing interim envy-freeness

For $i, j \in N$, $g \in I$, and $\mu \in \mathcal{M}$, define

$$A_{i,j,g;\mu} = \mathbf{1}\{\mu(i) = g\}(v_i(g) - v_i(\mu(j))). \quad (4)$$

Lemma 3 (Linear form of iEF). *A lottery q is interim envy-free if and only if, for every $i, j \in N$ and $g \in I$,*

$$\sum_{\mu \in \mathcal{M}} A_{i,j,g;\mu} q_\mu \geq 0. \quad (5)$$

The inequalities with $j = i$ are tautologies.

Proof. Fix i, j, g . Let

$$p_{ig} = \mathbb{P}_{\mu \sim q}[\mu(i) = g] = \sum_{\mu: \mu(i)=g} q_\mu.$$

If $p_{ig} = 0$, then every term in (5) is zero, because the indicator $\mathbf{1}\{\mu(i) = g\}$ vanishes whenever $q_\mu > 0$. Hence (5) is automatically true.

Assume now that $p_{ig} > 0$. Then (5) is equivalent to

$$\sum_{\mu: \mu(i)=g} q_\mu v_i(g) \geq \sum_{\mu: \mu(i)=g} q_\mu v_i(\mu(j)).$$

Dividing by p_{ig} gives

$$v_i(g) \geq \frac{1}{p_{ig}} \sum_{\mu: \mu(i)=g} q_\mu v_i(\mu(j)) = \mathbb{E}_{\mu \sim q} [v_i(\mu(j)) \mid \mu(i) = g],$$

which is exactly the iEF condition (1). Conversely, if (1) holds and $p_{ig} > 0$, multiplying by p_{ig} gives (5). The case $p_{ig} = 0$ was already handled.

Finally, when $j = i$ we have $\mu(j) = \mu(i) = g$ whenever $\mu(i) = g$, so every coefficient $A_{i,i,g;\mu}$ is zero. Thus those inequalities are tautological. $\square$

4 The exact exponential-size LP

For each matching $\mu \in \mathcal{M}$, define the coefficient

$$c_\mu = \text{avN}(\mu) = \left(\prod_{i \in N} v_i(\mu(i)) \right)^{1/n}. \quad (6)$$

Consider the following linear program, with one variable q_μ for each perfect matching $\mu \in \mathcal{M}$:

$$\text{maximize} \quad \sum_{\mu \in \mathcal{M}} c_\mu q_\mu \quad (7)$$

$$\text{subject to} \quad \sum_{\mu \in \mathcal{M}} q_\mu = 1, \quad (8)$$

$$\sum_{\mu \in \mathcal{M}} A_{i,j,g;\mu} q_\mu \geq 0 \quad \forall i, j \in N, g \in I, \quad (9)$$

$$q_\mu \geq 0 \quad \forall \mu \in \mathcal{M}. \quad (10)$$

One may omit the tautological inequalities with $j = i$.

Theorem 4 (Exactness of the LP). *The feasible solutions of (7)–(10) are exactly the iEF lotteries over perfect matchings. Consequently:*

- (i) *if the LP is infeasible, then no iEF lottery exists;*
- (ii) *if the LP is feasible, its optimal value is exactly*

$$\max_{q \text{ iEF}} \mathbb{E}_{\mu \sim q} [\text{avN}(\mu)];$$

- (iii) *every optimal LP solution is an optimal iEF lottery, and every optimal iEF lottery is an optimal LP solution with the same objective value.*

Proof. The equality (8) and the nonnegativity constraints (10) say exactly that q is a probability distribution over $\mathcal{M}$. By Lemma 3, the inequalities (9) say exactly that q is interim envy-free. Finally, by (3) and (6), the LP objective is exactly the expected average Nash welfare of the lottery q . Hence both the feasible region and the objective agree with the original optimization problem. $\square$

The theorem gives the following exact algorithm.

Step 1. Enumerate all $n!$ perfect matchings $\mu \in \mathcal{M}$.

Step 2. Compute c_μ for every matching μ .

Step 3. Compute $A_{i,j,g;\mu}$ for every triple (i, j, g) and every matching μ .

Step 4. Solve (7)–(10) exactly.

Step 5. If the LP is infeasible, report that no iEF lottery exists. Otherwise return an optimal solution q^* , represented by the matchings with $q_\mu^* > 0$ and their probabilities.

This is an exact finite algorithm, but it is exponential in n . The number of variables is

$$|\mathcal{M}| = n! = \exp(\Theta(n \log n)).$$

If the valuations are rational, then all constraints have rational coefficients. The objective coefficients c_μ are nonnegative real algebraic numbers of the form $r^{1/n}$ with rational $r \geq 0$. Therefore exact comparison of objective values at rational vertices can be carried out using standard arithmetic for real algebraic numbers. This observation justifies exactness, but it does not imply polynomial running time.

5 A polynomial support bound

Although the explicit LP has exponentially many variables, some optimal solution has polynomially bounded support.

Theorem 5 (Small-support optimum). *Suppose the LP (7)–(10) is feasible. Then there exists an optimal iEF lottery whose support has size at most*

$$1 + n^2(n - 1). \quad (11)$$

If the tautological constraints with $j = i$ are retained, the weaker bound $1 + n^3$ also holds.

Proof. Omit the tautological constraints with $j = i$. There is one equality constraint, namely (8), and there are $n^2(n - 1)$ remaining iEF inequalities. Introduce one nonnegative slack variable for each iEF inequality, writing each inequality as an equality. The resulting standard-form LP has

$$1 + n^2(n - 1)$$

equality constraints. Since the feasible region is contained in the simplex, the objective is bounded whenever the LP is feasible. Hence an optimum is attained at a basic feasible solution. In a basic feasible solution, at most $1 + n^2(n - 1)$ variables among the original variables and slack variables are basic, and every positive original probability variable q_μ must be basic. Therefore the number of positive matching probabilities is at most $1 + n^2(n - 1)$. $\square$

Remark 6 (What the support bound certifies). *The support bound gives a compact representation of some optimal lottery once such a lottery has been found. A list of supported matchings and probabilities can directly certify feasibility and objective value. By itself, however, such a list does not certify optimality; an optimality certificate would require additional dual information or another proof that no feasible lottery has larger expected average Nash welfare.*

6 The dual and the separation obstruction

The dual formulation clarifies why the exponential LP above does not resolve the polynomial-time computational question.

Put a nonnegative dual variable $y_{i,j,g} \geq 0$ on the inequality (9) and a free dual variable $z \in \mathbb{R}$ on the equality (8). Using the Lagrangian convention for a maximization problem with constraints of the form $Aq \geq 0$, the dual can be written as

$$\text{minimize } z \quad (12)$$

$$\text{subject to } z \geq c_\mu + \sum_{i,j \in N} \sum_{g \in I} y_{i,j,g} \mathbf{1}\{\mu(i) = g\} (v_i(g) - v_i(\mu(j))) \quad \forall \mu \in \mathcal{M}, \quad (13)$$

$$y_{i,j,g} \geq 0 \quad \forall i, j \in N, g \in I. \quad (14)$$

Indeed, for feasible primal q and dual (y, z) , multiplying (13) by q_μ and summing over μ gives

$$z \geq \sum_{\mu \in \mathcal{M}} c_\mu q_\mu + \sum_{i,j,g} y_{i,j,g} \sum_{\mu \in \mathcal{M}} A_{i,j,g;\mu} q_\mu \geq \sum_{\mu \in \mathcal{M}} c_\mu q_\mu,$$

so z is an upper bound on every feasible primal objective value. Since this is the usual finite-dimensional LP dual, strong duality holds whenever the primal is feasible.

A polynomial-time ellipsoid approach to the dual would require a polynomial-time separation oracle for (13). Given candidate dual variables (y, z) , such an oracle would have to decide whether there exists a perfect matching μ with

$$c_\mu + \sum_{i,j \in N} \sum_{g \in I} y_{i,j,g} \mathbf{1}\{\mu(i) = g\} (v_i(g) - v_i(\mu(j))) > z. \quad (15)$$

Equivalently, it would have to maximize over perfect matchings the expression

$$\left(\prod_{i \in N} v_i(\mu(i)) \right)^{1/n} + \sum_{i,j \in N} \sum_{g \in I} y_{i,j,g} \mathbf{1}\{\mu(i) = g\} (v_i(g) - v_i(\mu(j))). \quad (16)$$

The second term is the familiar edge-pair-weighted expression arising from the iEF constraints. The first term is the average Nash welfare of the matching. In contrast, the log-Nash objective for a deterministic matching is

$$\sum_{i \in N} \log v_i(\mu(i)),$$

which decomposes additively over the matched edges when all selected values are positive. That additive structure is exactly what makes the log-Nash case compatible with the known separation strategy. The geometric-mean term in (16) is not handled by the argument above.

Thus the exponential primal LP avoids the separation problem by enumerating every matching. It does not give a polynomial-time separation oracle, and it does not prove that such an oracle exists or cannot exist.

7 Relation to the authors' open problem

The open problem asks, in effect, for an algorithmic or complexity-theoretic understanding of the expected average Nash objective under iEF constraints. The formulation proved in Theorem 4 gives an algorithm only in the very literal sense that every finite LP can be solved after explicitly listing all its variables and constraints. Because there are $n!$ matching variables, this does not address the intended efficient-computation question.

Moreover, the existence of an exponential LP over matchings is not new relative to the authors' framework. The main challenge in their matching results is not writing the primal LP, but solving it in polynomial time through the dual and a separation oracle. The argument in this note does not supply the missing separation oracle for expected average Nash welfare and does not provide a hardness result for the separation problem (16).

Consequently, the correct interpretation is as follows.

Theorem 7 (Status relative to the open problem). *The LP formulation (7)–(10) is a correct exact exponential-size formulation of the expected average Nash iEF optimization problem. It yields a finite exact algorithm and the support bound of Theorem 5. It does not resolve the computational open problem of computing such lotteries efficiently, nor does it determine the problem's computational complexity.*

Proof. The correctness and exactness claims are precisely Theorem 4. The support statement is Theorem 5. The algorithm obtained from these results enumerates all $n!$ perfect matchings and hence is exponential in n . Finally, the displayed separation objective (16) identifies the problem that would have to be solved to turn the dual into a polynomial-time ellipsoid algorithm. No polynomial-time algorithm or hardness proof for that separation problem is supplied here. Therefore the efficient-computation and complexity-classification questions remain unaddressed by this note. $\square$

8 Conclusion

The proposed LP observation contains no serious error in its core mathematical equivalence: interim envy-freeness is linear after conditioning probabilities are cleared, and expected average Nash welfare is linear in the lottery distribution once deterministic matching values are treated as coefficients. The resulting LP is therefore exact.

The serious issue is one of interpretation. An exact LP with $n!$ variables is not the kind of result asked for in the open problem, because the authors' contribution for related objectives was precisely to avoid this exponential enumeration. The present note proves only an exact exponential formulation and a small-support consequence. It should not be described as a solution of the authors' open problem.

References

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